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SY3 A3 EXP 9
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FAQ

1. What is a tree?

Ans: A tree is a non-linear data structure.
It is used to store data in a hierarchical form.
It consists of nodes connected by edges.

2. What is a binary tree?

Ans: A binary tree is a tree in which each node has at most two children.
These are called the left subtree and right subtree.
It may also be an empty tree.

3. What are the types of Binary Tree?

Ans: The types of binary trees are:

1. Complete Binary Tree
2. Full Binary Tree
3. Binary Search Tree
4. Skewed Binary Tree
5. Threaded Binary Tree.

4. What are tree traversal techniques?

Ans: Tree traversal means visiting all nodes of a tree.
There are two types of traversal techniques:

1. Recursive
2. Non-recursive.

5. What are recursive tree traversal techniques?

Ans: There are three types of recursive tree traversal techniques:

1. Inorder – Left, Root, Right
2. Preorder – Root, Left, Right
3. Postorder – Left, Right, Root.

6. What are AVL trees?

Ans: AVL trees are height-balanced binary search trees.
The height difference between the left and right subtrees of each node is at most 1.
It helps in faster searching.

7. What are the basic operations that can be performed with a tree?

Ans: The basic operations performed on a tree are:

1. Creation
2. Searching
3. Insertion
4. Deletion
5. Traversal.

8. Explain the term degree and height of tree.

Ans: Degree of a node is the number of subtrees of that node.
Height of a tree is the maximum level of any node in the tree.
It is also called the depth of the tree.